

Data Detectives

Anuja Alluri, Makaela Sellers, Subiksha Vaidhyanathan, Varshitha Thanam

We wanted to create a machine learning program that would predict the location and type of crime committed next in Cincinnati.

We found a large dataset with crimes logged by the Cincinnati Police Department and preprocessed it—the first time—so it only had the type of crime, the GPS coordinates, and date. The first thing we did was try a linear regression model by uploading the dataset and pre-processing it and then training the model and testing it. We noticed in the output graph when we tested the data using linear regression was the plotted points overcrowded one another which made it difficult to differentiate between the various plotted points because of which we considered the model to be an example which was not a very successful as it didn't produce the expected results.

We decided that the linear regression model wasn't working as well as we wanted, so we tried doing a neural networking model. We chose to, instead of predicting the location, predict what the next offence would be. It ended up that we couldn't accurately predict the offence type, and so we decided to try a binary violent or non-violent offence prediction. We made a key that defined violent crimes, described by a document from the National Institute of Justice [1], as 1, and non-violent crimes as 0.

However, even after we did this, we were unable to accurately predict whether the next crime would be violent or non-violent. We would need a few more parameters that, due to time, we were unable to add. Such parameters include a categorization of GSP coordinates in certain districts that are very safe, moderately safe, moderately dangerous, and very dangerous—which

we would have looked for in the Cincinnati Police Department website, or what officers were on duty.

While neither of our models worked to achieve what we set out to accomplish, we believe we learned a lot about machine learning algorithms, and we would know what we'd do if we had enough time to get our model to work.

[1] <https://nij.ojp.gov/topics/crimes/violent-crime>